

SKILL 14

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    FILE *fp;
    FILE *errfile;
    char text[200];

    printf("=== Append Mode Test ===\n");

    fp = fopen("append.txt", "a");

    if (fp == NULL) {
        printf("Failed to open append.txt\n");
        return 1;
    }

    fprintf(fp, "New line added using append mode\n");
    fclose(fp);

    printf("Data appended successfully\n");

    printf("\n=== Verify Appended Data ===\n");

    fp = fopen("append.txt", "r");

    if (fp != NULL) {
        while (fgets(text, sizeof(text), fp) != NULL) {
            printf("%s", text);
        }

        fclose(fp);
    }
}
```

```

printf("\n=== stderr Redirection Test ===\n");

errfile = freopen("error.txt", "w", stderr);

if (errfile == NULL) {
    printf("Could not redirect stderr\n");
    return 1;
}

fprintf(stderr, "This is a test error message\n");

FILE *missing = fopen("missing_file.txt", "r");

if (missing == NULL) {
    perror("Error opening missing_file.txt");
}

fclose(stderr);

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printf("Error output saved to error.txt\n");

printf("\n=== Verify Error File ===\n");

fp = fopen("error.txt", "r");

if (fp != NULL) {
    while (fgets(text, sizeof(text), fp) != NULL) {
        printf("%s", text);
    }

    fclose(fp);
}

printf("\n=== Task 14 Complete ===\n");

return 0;
}

```

```

nishanth@LAPTOP-6Q3ON07H: ~/task14
nishanth@LAPTOP-6Q3ON07H:~$ mkdir task14
nishanth@LAPTOP-6Q3ON07H:~$ cd task14
nishanth@LAPTOP-6Q3ON07H:~/task14$ nano task14.c

```

```
nishanth@LAPTOP-6Q3ON07H:~/task14$ ./task14.py
=== Append Mode Test ===
Data appended successfully

=== Verify Appended Data ===
Old data already in file
New line added using append mode

=== stderr Redirection Test ===
Error output saved to error.txt

=== Verify Error File ===
This is a test error message
Error opening missing_file.txt: No such file or directory

Task 14 Complete
```

```
Task 14 Complete
nishanth@LAPTOP-6Q3ON07H:~/task14$ cat append.txt
Old data already in file
New line added using append mode
nishanth@LAPTOP-6Q3ON07H:~/task14$ cat error.txt
This is a test error message
Error opening missing_file.txt: No such file or directory
nishanth@LAPTOP-6Q3ON07H:~/task14$
```